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Poisson Equation and Quasi-Fermi Potentials

This chapter discusses the Poisson equation, electrostatic potential, and quasi-Fermi potentials.

In all semiconductor devices, mobile charges (electrons and holes) and immobile charges (ionized dopants or traps) play a central role. The charges determine the electrostatic potential and, in turn, are themselves affected by the electrostatic potential. Therefore, each electrical device simulation at the very least must compute the electrostatic potential.

When all contacts of a device are biased to the same voltage, the device is in equilibrium, and the electron and hole densities are described by a constant quasi-Fermi potential. Therefore, together with the electrostatic potential, the relation between the quasi-Fermi potentials and the electron and hole densities is sufficient to perform the simplest possible device simulation.

Maxwell–Boltzmann statistics is the default in Sentaurus Device, and you can choose to use Fermi–Dirac statistics as described in Fermi Statistics on page 259.

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Electrostatic Potential

The electrostatic potential is the solution of the Poisson equation, which is:

Equation 38

$$\nabla \cdot \left(\varepsilon \nabla \phi - \vec{P} \right) = -q\left(p - n + N_{\mathrm{D}} - N_{\mathrm{A}}\right) - \rho_{\mathrm{trap}}$$

where:

- ε is the electrical permittivity.
- $\vec{P}$ is the ferroelectric polarization (see Chapter 29, Ferroelectric Materials on page 1003).
- q is the elementary electronic charge.
- n and p are the electron and hole densities.
- N_D is the concentration of ionized donors.
- N_A is the concentration of ionized acceptors.
- ρ_{trap} is the charge density contributed by traps and fixed charges (see Chapter 17, Traps and Fixed Charges on page 608).

The dataset name for the electrostatic potential is `ElectrostaticPotential`. The right-hand side of Equation 38 (divided by $-q$) is stored in the `SpaceCharge` dataset. The keyword for the Poisson equation in the `Solve` section is `Poisson`.

The dimensionless relative permittivity (that is, the permittivity in units of the vacuum permittivity ε_0) is given by the parameter `epsilon` in the `Epsilon` parameter set. The boundary conditions for the Poisson equation are discussed in Chapter 10, Boundary Conditions on page 289.

Dipole Layer

At material interfaces, dipole layers of immobile charges can occur, leading to a potential jump across the interface. They are modeled by:

Equation 39

$$\phi_2 - \phi_1 = \frac{q\,\sigma_D}{\varepsilon_r\,\varepsilon_0}$$

where:

- ϕ_1 and ϕ_2 refer to the electrostatic potential at both sides of the interface (index 1 is the reference side).
- σ_D is the dipole surface density.
- ε_0 is the free space permittivity.
- ε_r is the relative interface permittivity.
- q is the elementary electronic charge.

Feedback

The dipole interface model can be invoked for insulator–insulator interfaces by specifying the following in the `Physics` section of the interface:

```
Dipole ( Reference = "R1" )
```

Here, `Reference` denotes the reference side of the interface, and it can be either a region name or the material name.

The parameters of the model are given in the `Dipole` parameter set of the region or material interface section in the parameter file, and are listed in Table 30.

Note:

At critical points where heterointerfaces and dipole interfaces intersect, semiconductor regions must be interface connected. Interface traps with tunneling cannot be located on dipole interfaces.

Table 30 Dipole model parameters

Symbol	Parameter name	Default value	Unit
σ_D	`sigma_D`	0	cm^{-1}
ε_r	`epsilon`	3.9	1

Equilibrium Solution

For specific models (see Modified Ohmic Contacts on page 291, Barrier Lowering at Schottky Contacts on page 296, Floating Semiconductor Contacts on page 314, and Conductive Insulators on page 335), the equilibrium electrostatic potential is required internally. In this case, the Poisson equation is solved with equilibrium boundary conditions before any other `Coupled` statement and the solution is saved for use during the simulation.

You can control the parameters of the nonlinear solver for the equilibrium Poisson equation. The parameters `Iterations`, `Digits`, `NotDamped`, `LineSearchDamping`, and `RelErrControl`, described in Table 354 on page 1863, can be specified as options to the keyword `EquilibriumSolution` in the `Math` section (see Table 224 on page 1750) to control the Newton solver behavior.

The following example forces the Newton solver to solve for an equilibrium solution with 7 digits as the target accuracy and a maximum number of 100 iterations:

```
Math {
    ...
    EquilibriumSolution(Iterations=100 Digits=7)
    ...
}
```

Quasi-Fermi Potential With Boltzmann Statistics

Electron and hole densities can be computed from the electron and hole quasi-Fermi potentials, and vice versa.

If Boltzmann statistics is assumed, the formulas read:

Equation 40

$$n = N_C \exp\left(\frac{E_{F,n} - E_C}{kT}\right)$$

Equation 41

$$p = N_V \exp\left(\frac{E_V - E_{F,p}}{kT}\right)$$

where:

- N_C and N_V are the effective density-of-states.
- $E_{F,n} = -q\Phi_n$ and $E_{F,p} = -q\Phi_p$ are the quasi-Fermi energies for electrons and holes.
- Φ_n and Φ_p are electron and hole quasi-Fermi potentials, respectively.
- E_C and E_V are conduction and valence band edges, defined as:

Equation 42

$$E_C = -\chi - q(\phi - \phi_{ref})$$

Equation 43

$$E_V = -\chi - E_{g,eff} - q(\phi - \phi_{ref})$$

where χ denotes the electron affinity, $E_{g,eff}$ is the effective band gap, and ϕ_{ref} is a constant reference potential (see below). The zero level for the quasi-Fermi potential agrees with the zero level for the voltages applied at the contacts.

In unipolar devices, such as MOSFETs, it is sometimes possible to assume that the value of quasi-Fermi potential for the minority carrier is constant in certain regions. In this case, the concentration of the minority carrier can be directly computed from Equation 40 or Equation 41. This strategy is applied in Sentaurus Device if one of the carriers (electron or hole) is not specified inside the `Coupled` statement of the `Solve` section. Sentaurus Device uses an approximation scheme to determine the constant value of the quasi-Fermi potential.

Note:

In many cases if avalanche generation is important, the one carrier approximation cannot be applied even for unipolar devices.

The datasets for $E_{F,n}$. Φ_n, $E_{F,p}$, and Φ_p are named `eQuasiFermiEnergy`, `eQuasiFermiPotential`, `hQuasiFermiEnergy`, and `hQuasiFermiPotential`, respectively.

The `ConstRefPot` parameter allows you to specify ϕ_{ref} explicitly. Otherwise, Sentaurus Device computes ϕ_{ref} from the vacuum level, using the following rules:

- If there is silicon in any simulated semiconductor structure, then the intrinsic Fermi level of silicon is selected as reference, $\phi_{ref} = \Phi_{intr}(\mathrm{Si})$.
- Otherwise, if any simulated device structure contains GaAs, then $\phi_{ref} = \Phi_{intr}(\mathrm{GaAs})$.
- In all other cases, Sentaurus Device selects the material with the smallest band gap (assuming a mole fraction of 0) and takes the value of its intrinsic Fermi level as ϕ_{ref}.

Fermi Statistics

For the equations presented in the previous section, Boltzmann statistics was assumed for electrons and holes. Physically more correct, Fermi (also called Fermi–Dirac) statistics can be used. Fermi statistics becomes important for high values of carrier densities, for example, $n > 1\times10^{19}$ cm^{-3} in the active regions of a silicon device.

For Fermi statistics, Equation 40 and for Equation 41 are replaced by:

Equation 44

$$n = N_C F_{1/2}\left(\frac{E_{F,n} - E_C}{kT}\right)$$

Equation 45

$$p = N_V F_{1/2}\left(\frac{E_V - E_{F,p}}{kT}\right)$$

where $F_{1/2}$ is the Fermi integral of order 1/2.

Alternatively, you can write these expressions as:

Equation 46

$$n = \gamma_n N_C \exp\left(\frac{E_{F,n} - E_C}{kT}\right)$$

Equation 47

$$p = \gamma_p N_\mathrm{V} \exp\left(\frac{E_\mathrm{V} - E_{\mathrm{F},p}}{kT}\right)$$

where γ_n and γ_p are the functions of η_n and η_p:

Equation 48

$$\gamma_n = \frac{n}{N_\mathrm{C}} \exp\left(-\eta_n\right)$$

Equation 49

$$\gamma_p = \frac{p}{N_\mathrm{V}} \exp\left(-\eta_p\right)$$

Equation 50

$$\eta_n = \frac{E_{\mathrm{F},n} - E_\mathrm{C}}{kT}$$

Equation 51

$$\eta_p = \frac{E_\mathrm{V} - E_{\mathrm{F},p}}{kT}$$

Using Fermi Statistics

To activate Fermi statistics, you must specify the keyword `Fermi` in the `Physics` section:

```
Physics {
   ...
   Fermi
}
```

By default, the Joyce-Dixon approximation [1] is used for the Fermi integral in Equation 44 and Equation 45. An alternative approach is available based on the Fukushima approximation for Fermi integrals [2], which may offer benefits in terms of accuracy and efficiency and can be enabled using the following command:

```
Physics {
   ...
   Fermi (withFukushima)
}
```

Note:

Fermi statistics can be activated only for the entire device. Region-specific or material-specific activation is not possible, and the keyword `Fermi` can be defined in any `Physics` section.

Initial Guess for Electrostatic Potential and Quasi-Fermi Potentials in Doping Wells

To determine an initial guess for the electrostatic potential and quasi-Fermi potentials, a device is divided into doping well regions, where a doping well region is a semiconductor region consisting of a set of connected semiconductor elements bounded by nonsemiconductor elements or by vacuum (a doping well region might contain several mesh semiconductor regions). Then, each doping well region is further subdivided into wells of n-type and p-type doping, such that p-n junctions (`DopingConcentration=0`) serve as dividers between wells. For doping wells with more than one contact, the wells are further subdivided, such that no well is associated with more than one contact. Every well is connected uniquely to a contact or it has no contact (floating well).

Note:

Sentaurus Device uses integers to enumerate all doping wells. You can visualize the indices of the doping wells by plotting the `DopingWells` field.

In wells with contacts, the quasi-Fermi potential of the majority carrier is set to the voltage on the contact associated with the well. For wells that have no contacts, the following equations define the quasi-Fermi potential for the majority carriers:

Equation 52

$$\Phi_p = k_{\mathrm{float}} V_{\max} + (1 - k_{\mathrm{float}}) V_{\min}$$

Equation 53

$$\Phi_n = (1 - k_{\mathrm{float}}) V_{\max} + k_{\mathrm{float}} V_{\min}$$

where:

- $V_{\min}$ and $V_{\max}$ are the minimum and maximum of the contact voltages of the semiconductor wells with contacts in the doping well region, where the contactless well under consideration is located.
- The coefficient k_{float} is an adjustable parameter with a default value of 0. To change the value of k_{float}, use the keyword `FloatCoef` in the `Physics` section.

If a well has a contact and it is the only well in the doping well region to which it belongs, then the quasi-Fermi potential of the minority carrier is set equal to the quasi-Fermi potential of the majority carrier. For all other wells, the quasi-Fermi potential of the minority carrier is set to $V_{\min}$ or $V_{\max}$ if the well is n-type or p-type, respectively, with $V_{\min}$ and $V_{\max}$ previously described.

The electrostatic potential in a well is set to the value of the quasi-Fermi potential of the majority carrier adjusted by the built-in potential, that is, the initial solution will obey charge neutrality in all semiconductor regions.

Regionwise Specification of Initial Quasi-Fermi Potentials

You can set initial quasi-Fermi potentials regionwise. This is especially useful in charge-coupled device (CCD) simulations, where a CCD cell or region must be initially in a certain state (usually, deep depletion). By specifying an initial quasi-Fermi potential, the region or set of regions can be brought to the proper state or states at the start of the simulation.

The initial quasi-Fermi potentials for electrons and holes can be specified regionwise in the `Physics` section using `eQuasiFermi` for electrons and `hQuasiFermi` for holes, followed by the value in volts:

```
Physics(Region="region_1") {
   eQuasiFermi = 10
}
```

The initial quasi-Fermi potential is recomputed when the continuity equation for the respective carrier is solved. If the continuity equation for the carrier whose quasi-Fermi potential has been specified is not solved, then the quasi-Fermi potential does not change. In this case, the device can be biased to an initial state through a quasistationary or transient simulation, while keeping the initial specified quasi-Fermi potential.

Alternative Initial Guess for Quasi-Fermi Potentials From Linear Conduction Equation

Sentaurus Device provides an alternative method to initialize the quasi-Fermi potentials in semiconductor materials. It is more computationally intensive than their initialization in doping wells, but often gives better results, especially in complicated device structures with multiple contacts per doping well region.

In this approach, a solution of the linear conduction equations:

Equation 54

$$-\nabla \cdot \left(\sigma_{n_0} \nabla \Phi_n \right) = 0$$

Equation 55

$$-\nabla \cdot \left(\sigma_{p_0} \nabla \Phi_p \right) = 0$$

serves as an initial guess for the electron and hole quasi-Fermi potentials, Φ_n and Φ_p, respectively.

The conductivity coefficients for electrons (σ_{n_0}) and holes (σ_{p_0}) are approximated as:

Equation 56

$$\sigma_{n_0} = q \mu_{L,n} n_0$$

Equation 57

$$\sigma_{p_0} = q\,\mu_{L,p} p_0$$

where n_0, p_0 are the equilibrium carrier concentrations, and $\mu_{L,n}$, $\mu_{L,p}$ are the constant carrier mobilities. The concentrations n_0 and p_0 are calculated by invoking the charge neutrality condition (Equation 103 on page 290) and the law of mass action (Equation 104 on page 290). The mobilities $\mu_{L,n}$ and $\mu_{L,p}$ are taken as the values of the `mumax` parameter for electrons and holes, respectively, from the `ConstantMobility` parameter set (see Mobility due to Phonon Scattering on page 431).

The following boundary conditions are used when solving Equation 54 and Equation 55:

- At contacts, the Dirichlet boundary condition is applied, by setting the quasi-Fermi potentials equal to the voltage on the contact.
- For interfaces between a semiconductor and an insulator, a homogeneous Neumann boundary condition is imposed.
- At semiconductor–semiconductor and semiconductor–metal interfaces, the continuity of quasi-Fermi potentials is enforced.

In the command file, the initialization of quasi-Fermi potentials from the solution of Equation 54 and Equation 55 is activated by the `IniteQuasiFermi` and `InithQuasiFermi` options, respectively, set in the `Physics` section. You can use these options independently and can also use them in regionwise or materialwise `Physics` sections. The following example activates the linear conduction equation for the electron quasi-Fermi potential in region `n-epi`:

```
Physics (Region="n-epi") {
   IniteQuasiFermi
}
```

For interior interfaces where the linear conduction equation is solved only in one of the regions of the interface, the continuity condition for quasi-Fermi potentials is evaluated based on the initial guess coming from doping wells (see Initial Guess for Electrostatic Potential and Quasi-Fermi Potentials in Doping Wells on page 261). In metal regions, Equation 54 and Equation 55 are never solved. Instead, the metal conductivity equation is used to compute the Fermi level (see Transport in Metals on page 329).

Therefore, the linear conduction equation is solved by default only in semiconductor regions. However, the solution of the linear conduction equation can be extended to insulator regions by using the `includingInsulator` option. For example:

```
IniteQuasiFermi(includingInsulator)
InithQuasiFermi(includingInsulator)
```

In insulator regions, the conductivity coefficients and σ_{n_0} and σ_{p_0}, in Equation 54 and Equation 55, are approximated as $\sigma_{n_0} = \sigma_{p_0} = 1 / \rho_0$, where ρ_0 is the resistivity value of the corresponding insulator material. The resistivity value ρ_0 is taken as the value of the `Resist0` parameter from the `Resistivity` parameter set (see Transport in Metals on page 329).

Alternatively, for Equation 54 and Equation 55, only with the assumption that $\Phi_n = \Phi_p = \Phi$, the total linear conductance equation can be solved:

$$-\nabla \cdot \left(\left(\sigma_{n_0} + \sigma_{p_0} \right) \nabla \Phi \right) = 0$$

To get the solution of Equation 57 for the quasi-Fermi potentials Φ_n or Φ_p, you must specify `IniteQuasiFermi(TotalConductance)` or `InithQuasiFermi(TotalConductance)`, respectively.

In the presence of floating doping wells (FDWs) in the device structure (with no direct electrical contact to such a doping well), additionally, an FDW model keeps the quasi-Fermi potential based on the solution of the linear conductance equation in FDWs only, but the quasi-Fermi potential in other device areas are defined by Initial Guess for Electrostatic Potential and Quasi-Fermi Potentials in Doping Wells on page 261. This model is based on Equation 57 with an additional fitting parameter to control the surface conductance in the surface vicinity of the FDWs. To activate such a solution for electron (donor) or hole (acceptor) FDWs, `IniteQuasiFermi(FDW)` or `InithQuasiFermi(FDW)` should be used in `Physics`, correspondingly. To change a default surface conductance used with this FDW model, you must modify a surface conductance factor in `Math{ FDW_SurfCondFactor = <value> }`. The default value is equal to 1.

Note:

> The FDW model based on the linear conductance equation and solved together with the Poisson-only equation cannot replace the full drift-diffusion solution completely. It gives an approximate solution for the electrostatic potential with the assumption of unchanged conductivity during bias ramping.

If the continuity equation for the carrier whose quasi-Fermi potential has been specified is not solved, then the initial quasi-Fermi potential is recomputed for every ramping step. In this case, the concentration of the carrier is calculated directly from Equation 40 or Equation 41 (see Quasi-Fermi Potential With Boltzmann Statistics on page 258).

You can access the datasets storing the values of Φ_n, σ_{n_0}, Φ_p, and σ_{p_0} by using the keywords `eQuasiFermiPotential`, `eInitialConductivity/Element`, `hQuasiFermiPotential`, and `eInitialConductivity/Element`, respectively.

Electrode Charge Calculation

Sentaurus Device outputs the electrode charge to the current plot file (`*.plt` file) after each bias or time point.

For an electrode that contacts an insulator region only, the charge is computed from Gauss's law and represents the charge that would sit on the surface of a real contact to the device.

For an electrode that contacts a semiconductor region, the charge also is computed from Gauss's law; however, the Gaussian surface used for the integration includes the doping well associated with the electrode. In this case, the electrode charge represents the charge that sits on the electrode plus the space charge in the doping well. With `-ContactChargeIsWellCharge` in the global `Math` section, the well integration is suppressed for all contacts, so that the contact charge only accounts for the charge at the contact itself.

References

[1] W. B. Joyce and R. W. Dixon, "Analytic approximations for the Fermi energy of an ideal Fermi gas." *Applied Physics Letters*, vol. 31, no. 5, pp. 354–356, 1977.

[2] T. Fukushima, "Precise and fast computation of Fermi-Dirac integral of integer and half integer order by piecewise minimax rational approximation," *Applied Mathematics and Computation*, vol. 259, pp. 708–729, May 2015.